

BLOOD SYNC

PLATFORM

An AI-powered healthcare platform designed to streamline donor outreach and optimize blood inventory management through predictive insights.

Role

UI/UX Designer



INTRODUCTION

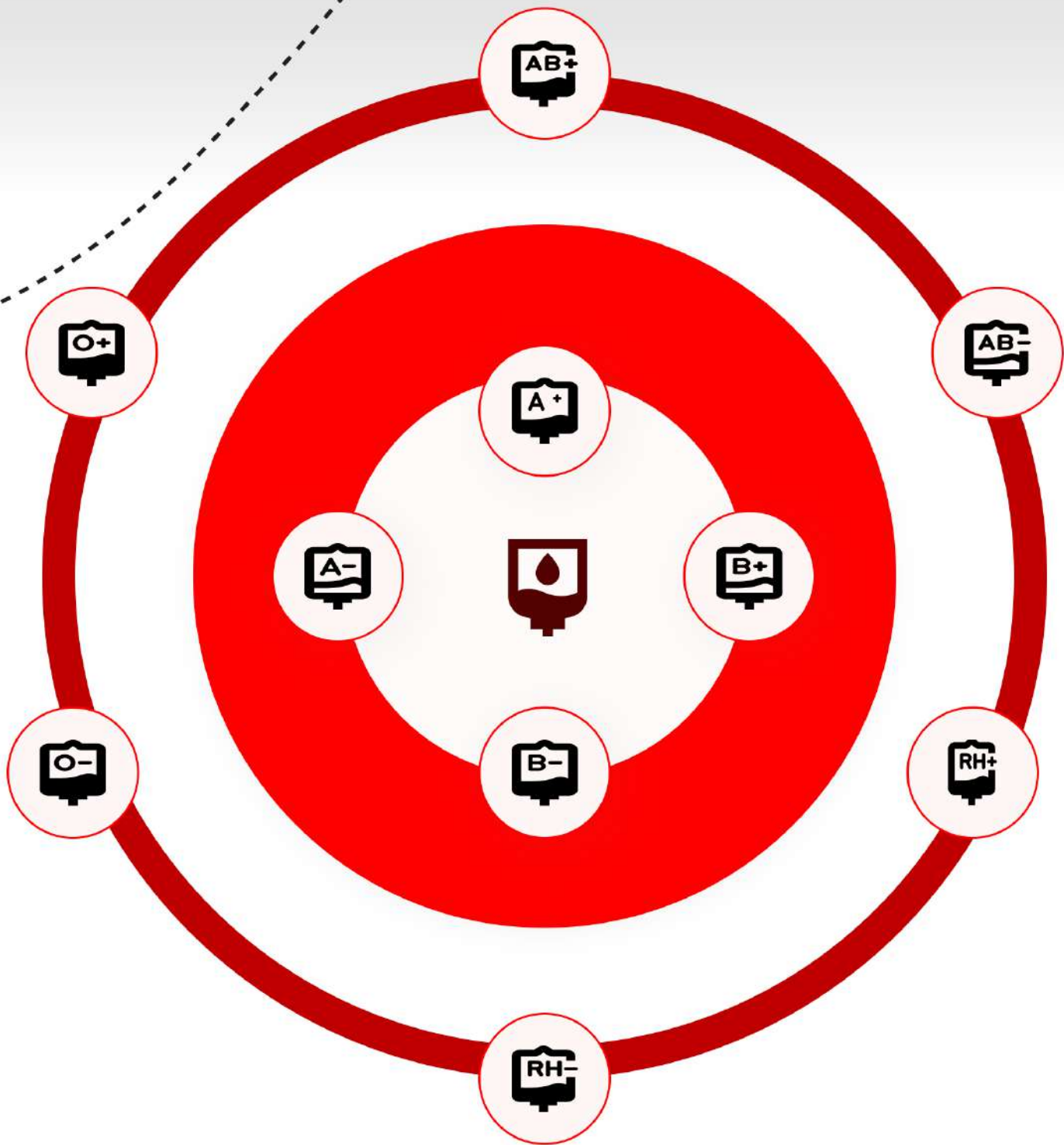
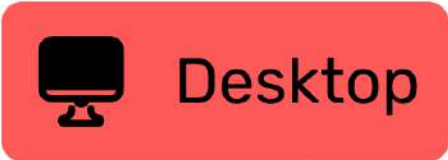
About This Project

This project functional UX and UI Designer to overcome the immense difficulty, broad minded and interactive and advanced UI Design.

GET STARTED

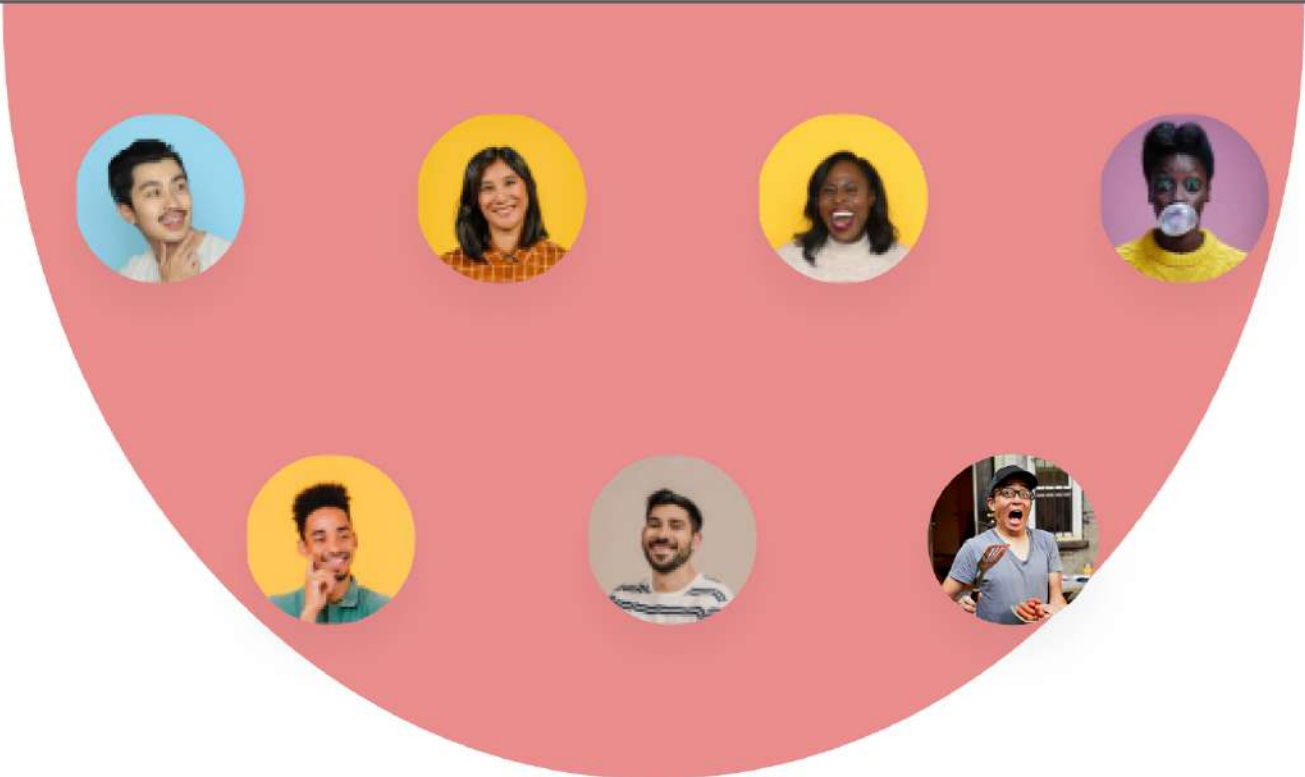
Overview

This platform is designed to improve blood management workflows for hospitals and blood banks by replacing manual processes with a centralized digital solution. It features intelligent donor prediction and real time blood inventory monitoring to support faster and more efficient decision making during emergencies.



The Story

The reliance on manual donor outreach and delayed blood inventory tracking often slows emergency response in healthcare. This inspired the design of BloodSync, a centralized platform that helps hospitals and blood banks manage donor coordination and blood availability more efficiently.



PROBLEM

Current blood management workflows in many hospitals and blood banks still rely on manual coordination, making emergency response slow and inefficient.

- Blood banks manually contact a large number of donors with uncertain response rates.
- Hospitals often identify blood shortages only after inventory reaches critical levels.
- Communication between hospitals and blood banks is fragmented and time-consuming.
- Donor availability and eligibility are difficult to track in real time..
- Critical operational data is scattered across multiple systems..
- Delayed coordination increases response time during medical emergencies.
- Lack of centralized insights makes proactive blood management challenging.



VS

SOLUTION

BloodSync provides a centralized healthcare platform that streamlines donor management and blood inventory monitoring through an intuitive dashboard.

- ✓ Intelligent donor prediction identifies the top 8–10 donors most likely to respond quickly.
- ✓ Real-time blood stock monitoring alerts hospitals when inventory falls below predefined thresholds.
- ✓ Centralized dashboards provide hospitals and blood banks with actionable operational insights.
- ✓ Interactive field-based analytics simplify monitoring and decision making.
- ✓ Secure role-based access ensures data is available only to authorized healthcare organizations.
- ✓ Visual reports and alerts help users prioritize critical actions efficiently.
- ✓ A unified platform improves coordination, reduces response time, and supports proactive blood management.

DESIGN PROCESS



Discover

Understand user needs, business goals, and existing healthcare workflows.



Define

Transform research insights into a clear product strategy and user experience.



Design

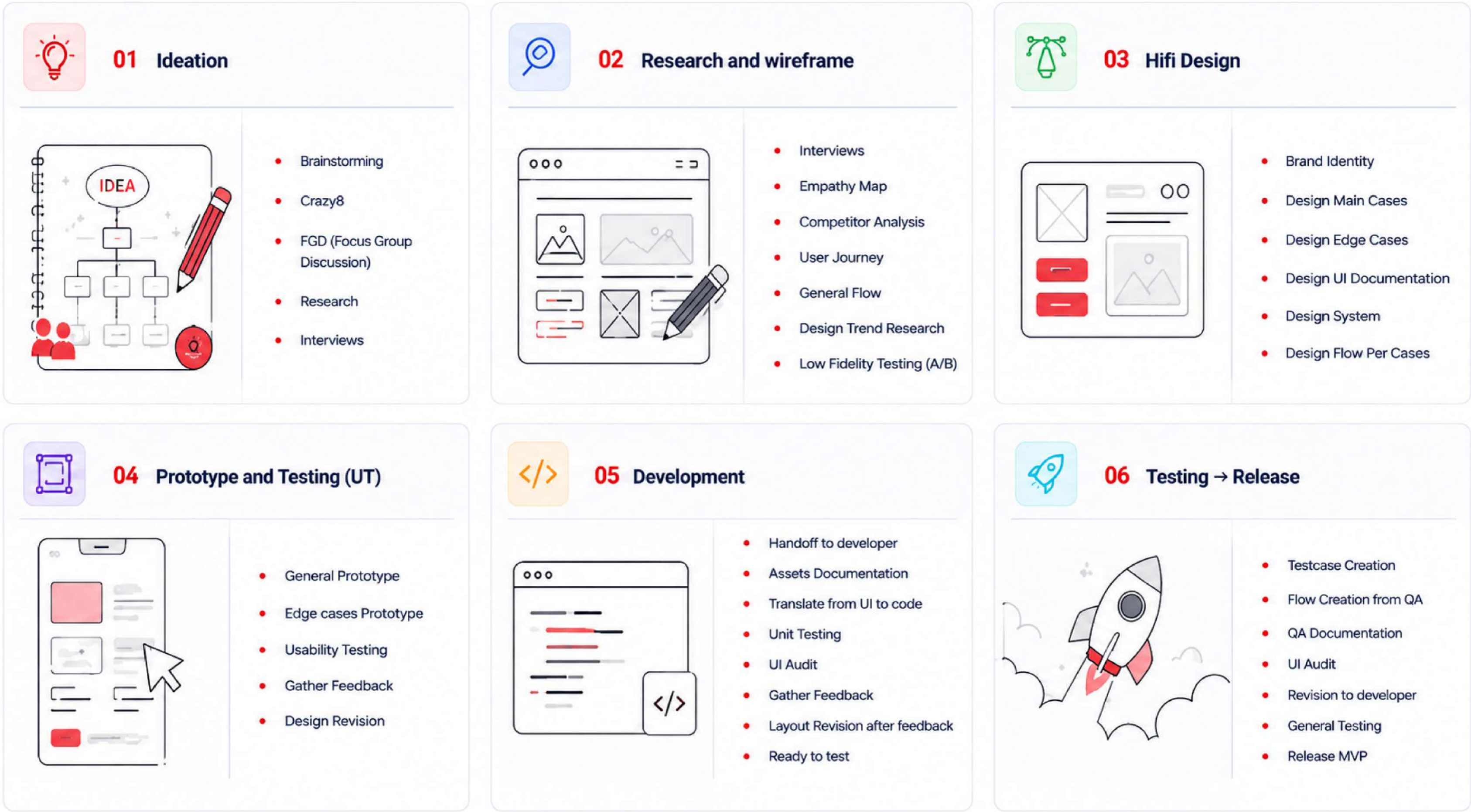
Create intuitive wireframes, a scalable design system, and high-fidelity interfaces.



Validate

Refine the experience through prototyping, usability testing, and iterative improvements.

Key Design Activities



My Role

As a UI/UX designer, I make sure that my designs are well received by users, and can be implemented by developers.



UI Documentation

creation of ui documentation so that developers can code well.



Provide Assets

ensure Assets contained in the UI are usable and provide in multiple sizes.



User testing

User testing using UT and AB testing, so that the App can be well received by the user.



UI Audit

Perform UI audits. make sure nothing goes wrong when the App is made.



Flow Documentation

when the application is continued, it can be easily carried out.



Research

This process is made so that the requirements and problems obtained meet the target market.

Design Strategy

A design strategy shows the value your product and services will bring to people and describes this value to a goal. It also describes about the steps you will take to achieve the goal.

Intention

“ To design a secure and user-friendly platform that streamlines blood management by helping hospitals and blood banks coordinate donor outreach and monitor blood inventory efficiently during critical situations.

Target User

- Hospital Administrators
- Blood Bank Staff
- Medical Coordinators

Design Constraint

- “
- Sensitive healthcare data requires secure access.
 - Interface must remain simple despite complex workflows.
 - Dashboard should display real-time information clearly.
 - Responsive design for desktop and tablet use.
 - High accessibility and readability for medical environments.

General task

- Sensitive healthcare data requires secure access.
- Interface must remain simple despite complex workflows.
- Dashboard should display real-time information clearly.
- Responsive design for desktop and tablet use.
- High accessibility and readability for medical environments.

Success Factor

- Faster donor outreach during emergencies
- Early detection of low blood stock
- Improved coordination between hospitals and blood banks
- Reduced manual effort in blood management
- Faster and more informed decision-making

USER RESEARCH

Research Plan

- Survey
- Interview
- Personas
- Empathy Map
- User Journey

User research is a process to find out the user point of view, their frustations and their struggle to get their understandig. I have two used methods which contributed hugely in aditio to my own findings

SURVEY

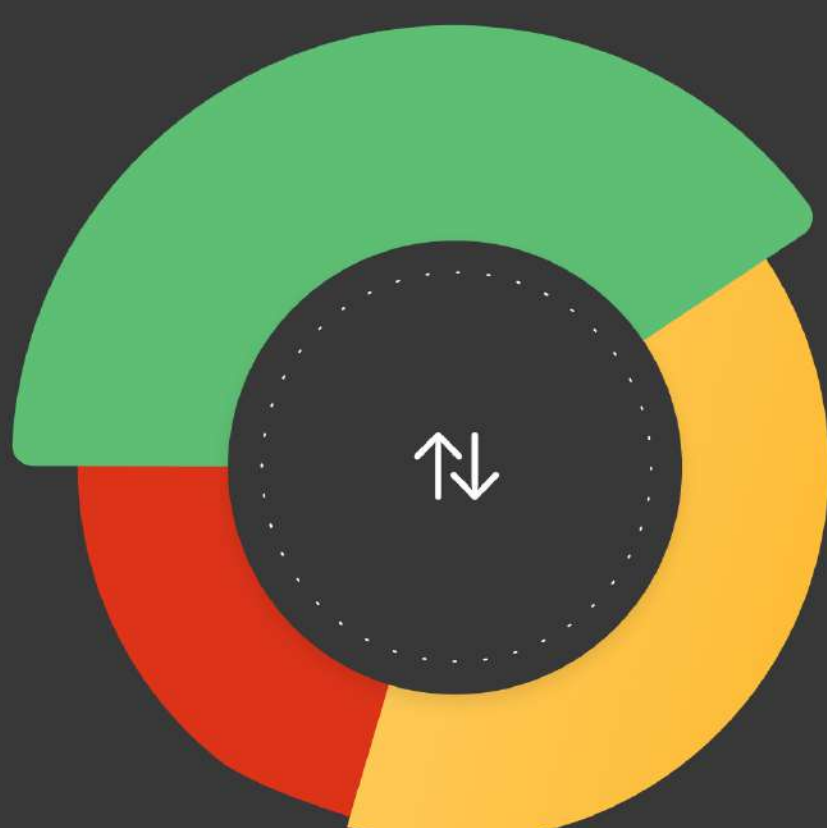
To better understand the challenges in blood management workflows, I conducted conceptual user research based on healthcare workflows, existing literature, and discussions with potential users. The research focused on identifying pain points experienced by hospitals and blood banks during donor management and blood inventory tracking.

“ How do you currently identify blood donors during emergencies?



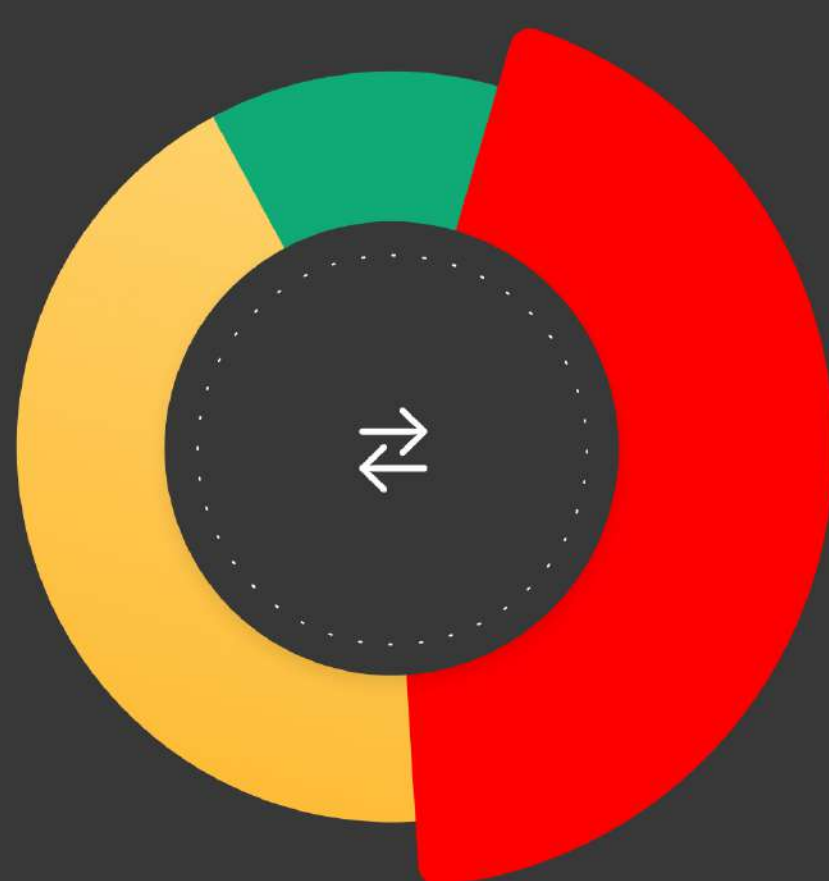
- Existing Donor Database - 50%
- Phone Calls - 30%
- Manual Records - 20%

“ What is the biggest challenge in donor management?



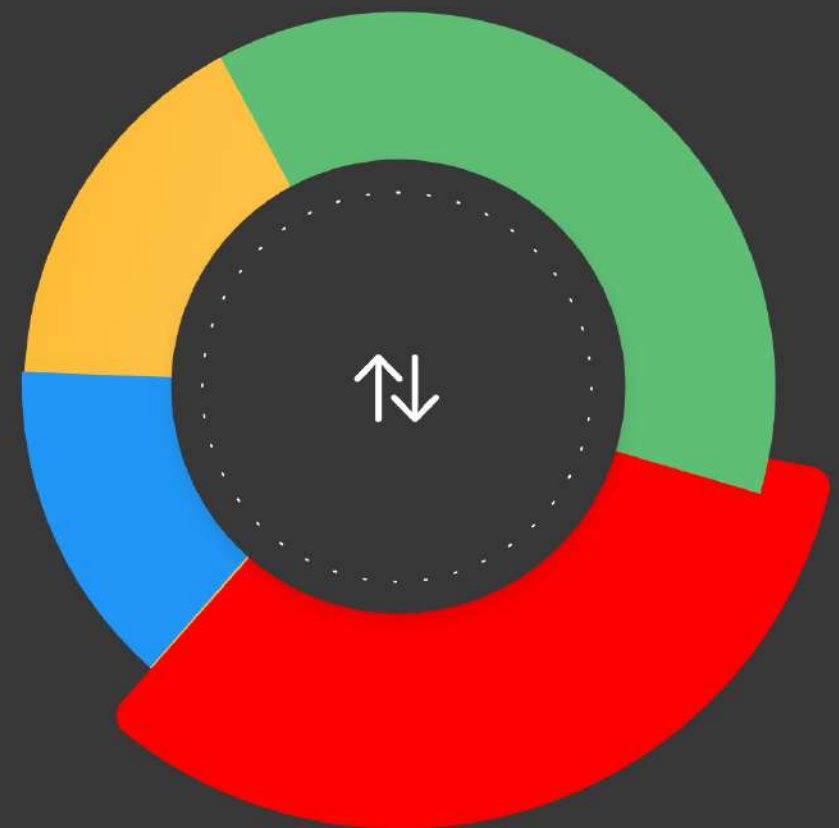
- Finding available donors quickly - 48%
- Low donor response rate - 32%
- Outdated donor records - 20%

“ How often do blood shortages impact operations?



- Frequently - 45%
- Occasionally - 40%
- Rarely - 15%

“ Which feature would provide the most value?



- AI Donor Prediction - 42%
- Blood Stock Alerts - 33%
- Analytics Dashboard - 15%
- Centralized Management - 10%

INTERVIEW



”

Dr.Aravind Kumar
Emergency Physician

"During emergency cases, we don't have time to call every donor one by one. If the system itself suggests the most likely donors, it will save valuable time."

”

Priya Lakshmi
Blood Bank Officer

"Sometimes we realize blood stock is getting low only at the last minute. An early alert will help us plan much better."

”

Suresh Kumar
Hospital Administrator

"Managing donor records manually is becoming difficult. We need everything in one place so our staff can work faster."

”

Divya Ramesh
Medical Coordinator

"If donor details, blood stock, and alerts are available in one dashboard, our team can make quicker decisions during emergencies."

”

Dr.Karthikeyan
Chicken seller

"Many donors are willing to donate, but finding the right person at the right time is always a challenge. A prediction system would be really helpful."

”

Nandhini
Hospital Operation Managaer

"A simple dashboard with clear alerts is much easier for our staff than checking multiple spreadsheets and records."

📌 Define

USER PERSONA



Full Name : Dr.Aravind Kumar
Age : 38 Years old
Occupation : Hospital Administrator
Location : Chennai, Tamil Nadu

Statement

"I need a simple and reliable system that helps me monitor blood availability and respond quickly during emergencies."

About

Dr. Aravind Kumar is an experienced Hospital Administrator responsible for managing daily hospital operations, including blood inventory coordination. During emergencies, he must ensure blood is available without delays. He prefers digital solutions that simplify complex workflows and provide quick access to critical information, enabling faster decisions and better patient care.

Goals

- Monitor blood inventory in real time.
- Receive early alerts for low blood stock.
- Coordinate efficiently with nearby blood banks.
- Reduce delays during emergency situations.
- Improve operational efficiency through a centralized dashboard.

Personality

- Responsible
- Organized
- Detail-Oriented
- Calm Under Pressure
- Analytical
- Tech-Savvy

Frustrations

- Manual tracking of blood inventory.
- Late notifications about blood shortages.
- Time-consuming coordination with blood banks.
- Managing information across multiple systems.
- Lack of real-time visibility into blood availability.

Empathy Map

An empathy map is a collaborative visualization used to articulate what we know about a particular type of user. It helps to synthesize research data to better assist in understanding how people make decisions.

Think & feel

"Will we have enough blood units for the next emergency?"

"I need to know about shortages before they become critical."

"Can we find suitable donors quickly during emergencies?"

"Managing blood inventory manually is stressful."

"We need O-negative blood immediately."

Blood inventory changing throughout the day.

"Blood stock is running low."

I need accurate information to make quick decisions."

"A centralized dashboard would reduce operational pressure."

Emergency blood requests from different departments.

"The blood bank is checking availability."

Manual records and spreadsheets.

What are

Hear

See

"Please arrange blood as soon as possible."

Delayed communication with blood banks.

"Please arrange blood as soon as possible."

Multiple systems for tracking inventory.

Monitor blood inventory regularly.

Check dashboard alerts throughout the day.

Ensure blood availability before emergencies escalate.

Contact nearby blood banks during shortages.

Review inventory reports and analytics.

Say and Do



Gain

- Early shortage alerts
- Faster donor coordination
- Centralized healthcare dashboard
- Better emergency preparedness
- Improved operational efficiency

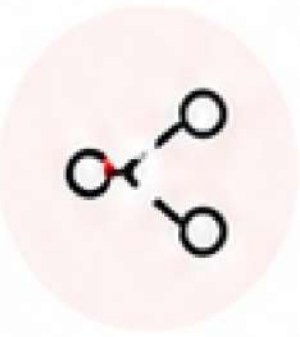


Pain

- Manual inventory tracking
- Delayed shortage identification
- Slow donor coordination
- Fragmented healthcare systems

RESEARCH INSIGHTS

Research showed that users struggle with blood-stock visibility, donor identification, request tracking, missed alerts, and understanding data. These findings helped define BloodSync’s key opportunities: real-time inventory, AI donor prediction, centralized requests, smart alerts, and actionable analytics.

WHAT WE HEARD	INSIGHT	OPPORTUNITY FOR BLOODSYNC
01 “ We don’t always know which blood groups are running low until it’s almost critical.” — BLOOD BANK OFFICER	 Visibility is reactive, not proactive. Users only discover low stock when it’s too late, leading to delayed actions.	 Real-time inventory dashboard Surface critical stock, trends and availability at a glance to enable early action.
02 “ Finding the right donors takes time. We call many, but only a few respond.” — HOSPITAL STAFF	 Donor discovery is manual and inefficient. Time is wasted reaching out to unavailable donors.	 AI-powered donor prediction Identify and prioritize likely responders to save time and increase response rates.
03 “ Requests come in from multiple places. It’s hard to keep track sometimes.” — HOSPITAL COORDINATOR	 Request tracking is fragmented. Information is scattered across calls, messages and sheets.	 Centralized request management Track, manage and communicate all blood requests in one organized workspace.
04 “ Important alerts often get missed in the middle of busy work.” — BLOOD BANK MANAGER	 Critical updates are easy to miss. Users need timely and clear alerts that demand attention.	 Smart notifications & alerts Deliver real-time, role-based alerts for critical stock, requests and expiring inventory.
05 “ We have data, but it’s hard to understand what to do next.” — HOSPITAL ADMINISTRATOR	 Data exists, but not actionable. Users need simple insights that guide confident and faster decisions.	 Actionable analytics Turn data into clear insights and recommendations that help users decide with confidence.

RESEARCH → IMPACT



01
Proactive operations

Users can act before shortages become critical.



02
Smarter outreach

AI helps focus on the right donors, improving response rates.



03
Streamlined workflows

Centralized information reduces back-and-forth and manual effort.



04
Faster response

Critical alerts ensure quick action when it matters most.



05
Better decisions

Clear insights empower users to make confident, data-driven decisions.

DESIGN SYSTEM

COLORS



TYPOGRAPHY

Inter

Aa

H1 / 32px / Bold
The quick brown fox

H2 / 24px / Semi Bold
The quick brown fox

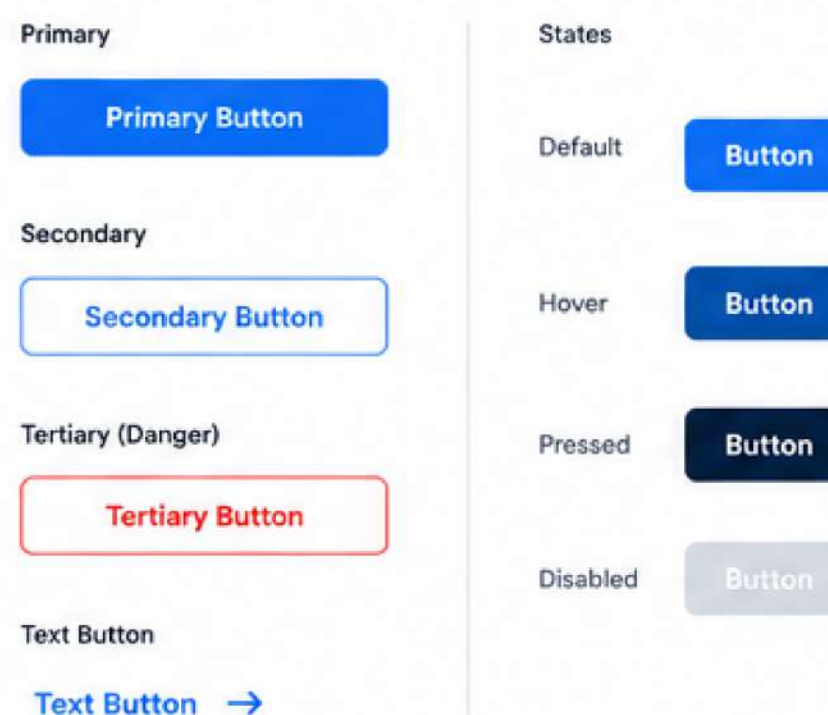
H3 / 20px / Semi Bold
The quick brown fox

Body / 16px / Regular
The quick brown fox

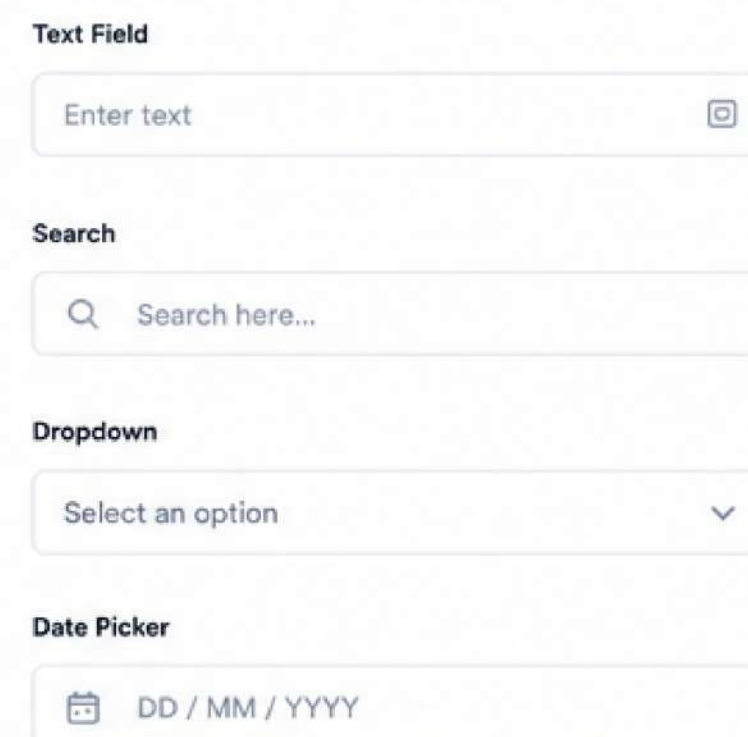
Small / 14px / Regular
The quick brown fox

Caption / 12px / Regular
The quick brown fox

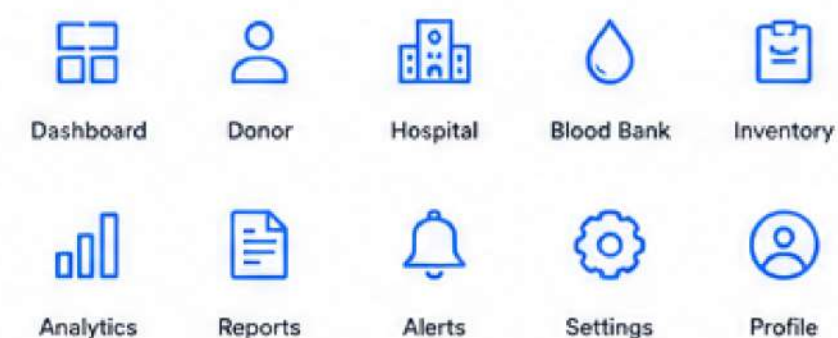
BUTTONS



INPUTS

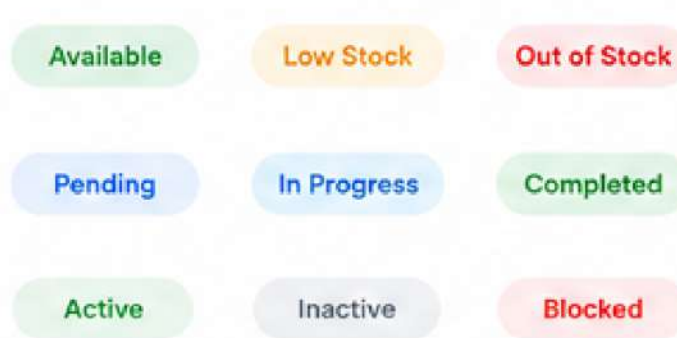


ICONS



All icons are 24px, Line style, 2px stroke, Rounded corners.

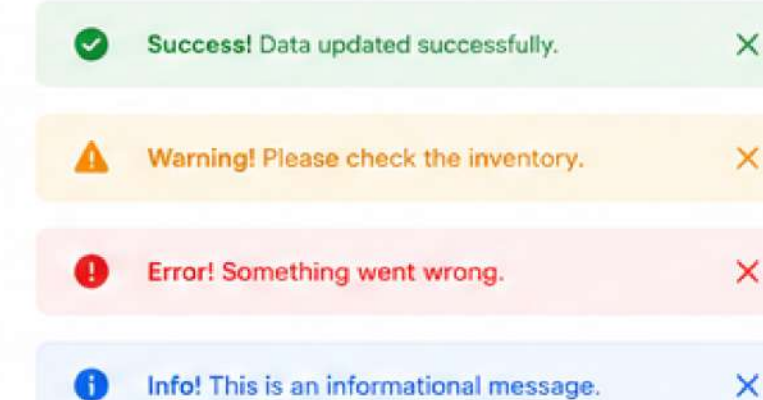
BADGES / STATUS



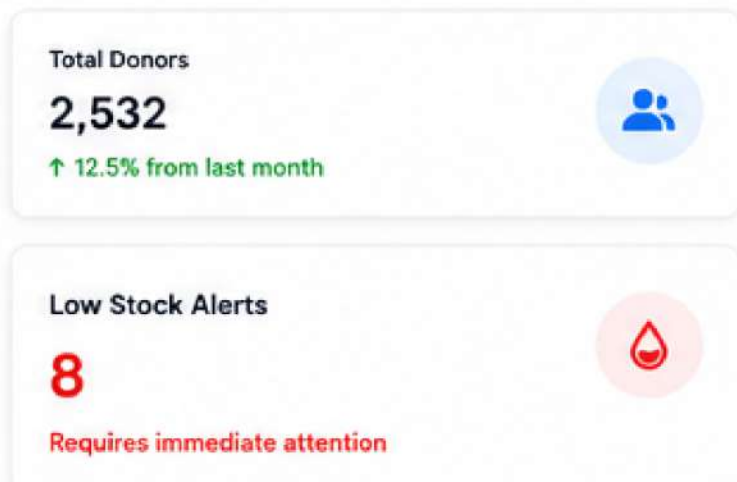
TABLE

Donor Name	Blood Group	Last Donation	Status
Rahul Sharma	O+	12 May 2024	Available
Priya Nair	A+	08 May 2024	Available
Arun Kumar	B+	30 Apr 2024	Pending
Sneha Patel	AB+	20 Apr 2024	Unavailable

ALERTS



CARDS



CHARTS

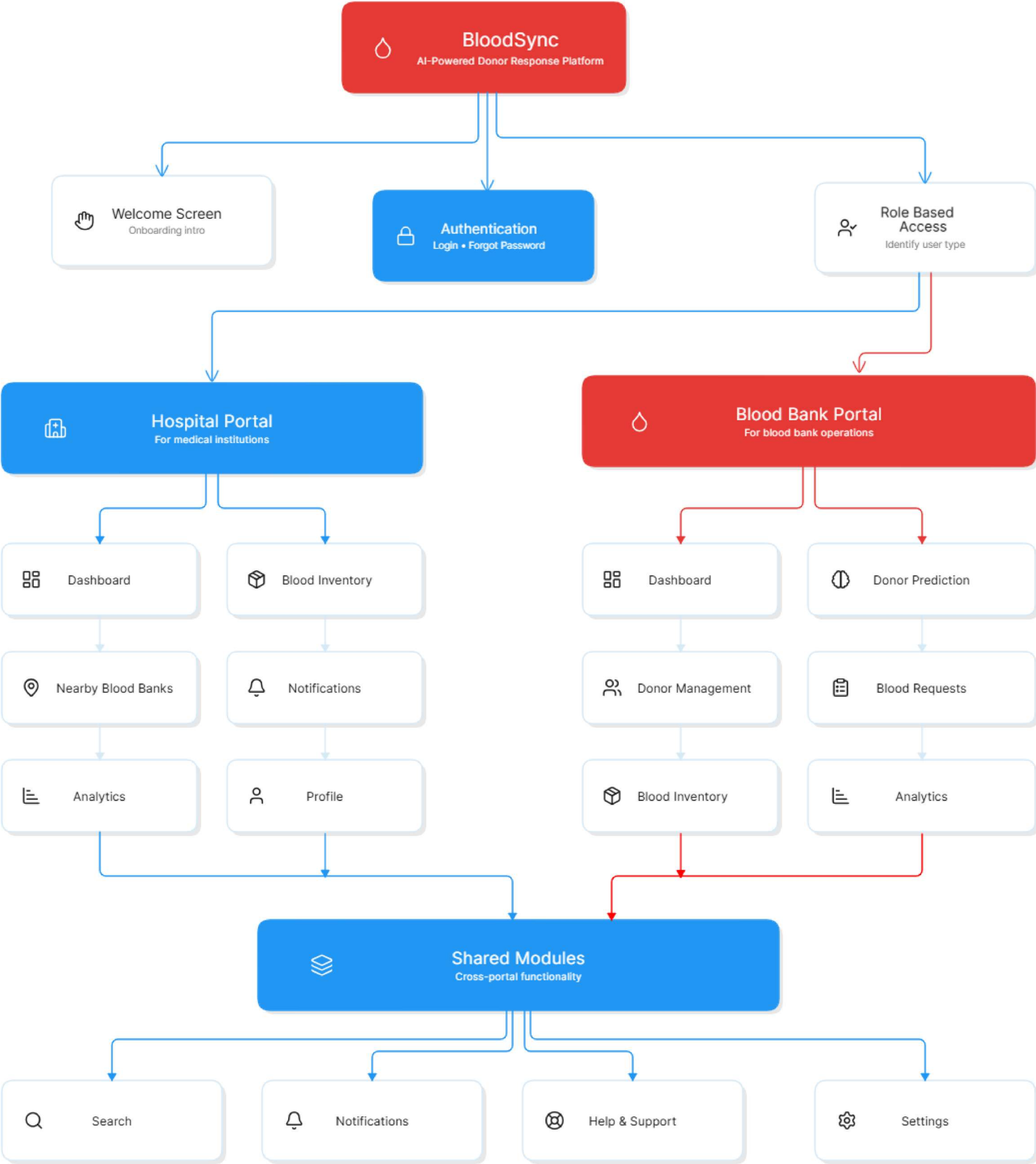


SHADOWS & SPACING



Information Architecture

The platform is structured around two primary user roles—Hospitals and Blood Banks. After authentication, users are directed to role-specific dashboards, ensuring quick access to relevant workflows while maintaining a consistent navigation structure.

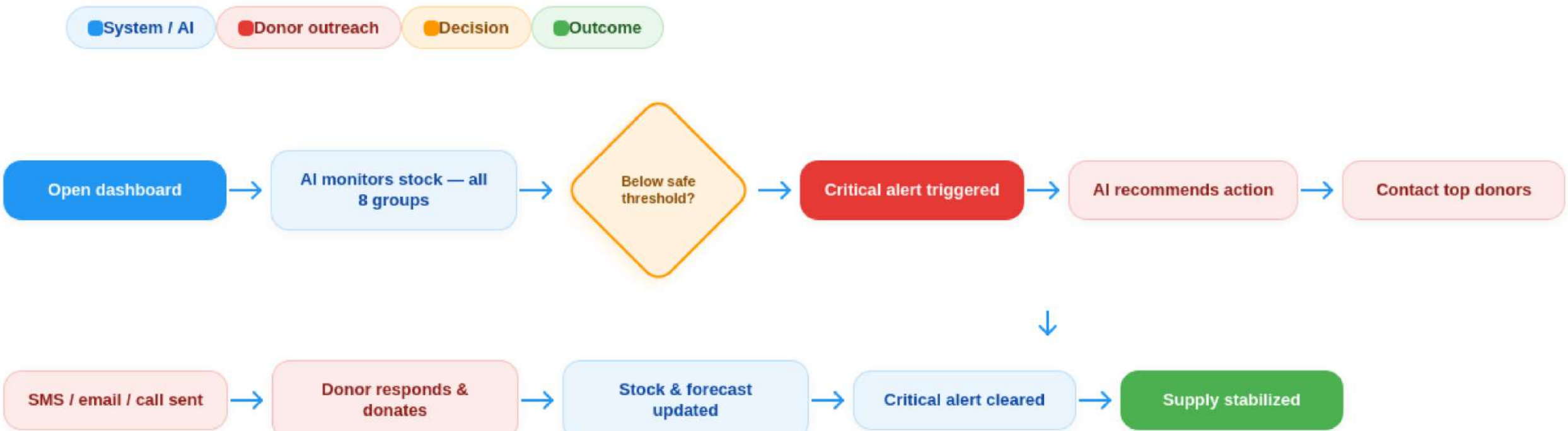


User Flow

The user flows illustrate how hospitals and blood banks navigate BloodSync to manage blood inventory, coordinate donor outreach, and respond efficiently to critical shortages.

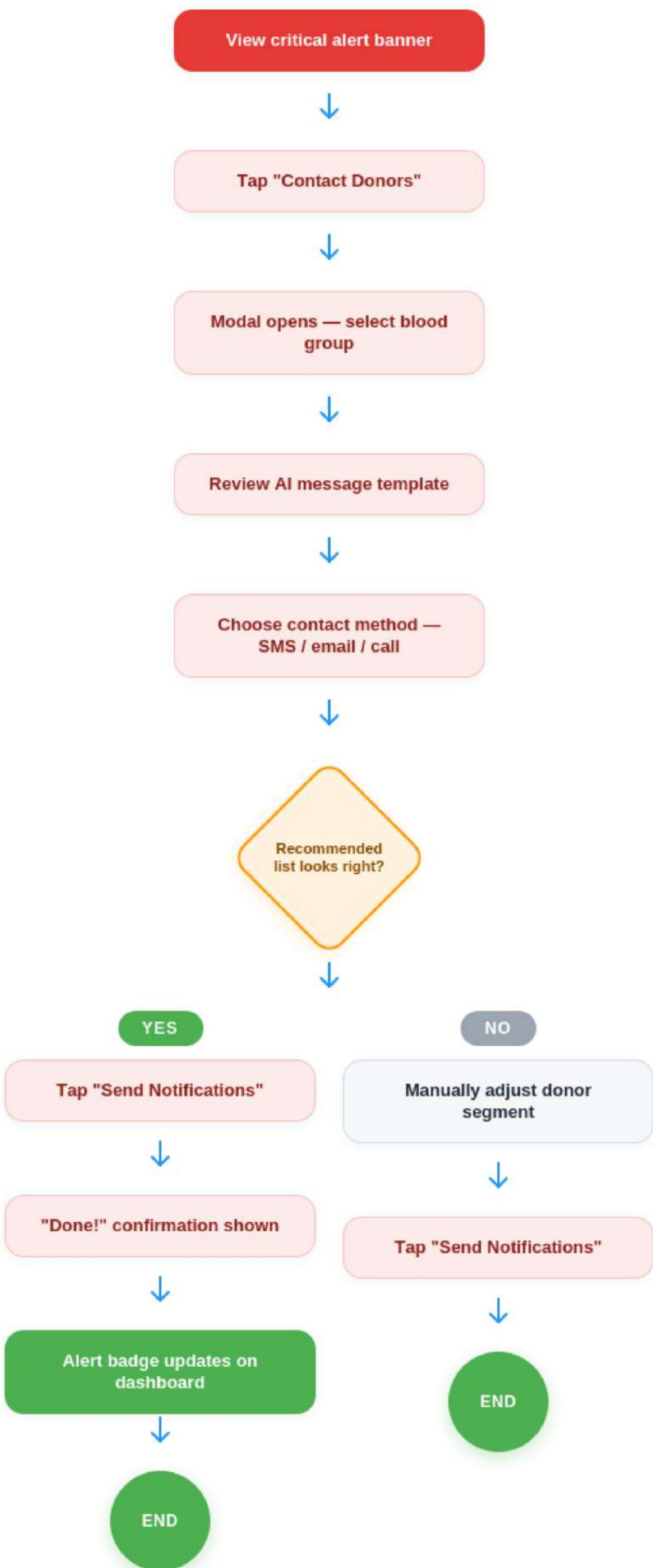
1 Overall system flow

How the AI model continuously monitors stock across all 8 blood groups and drives Dr. Sharma to a resolved shortage.



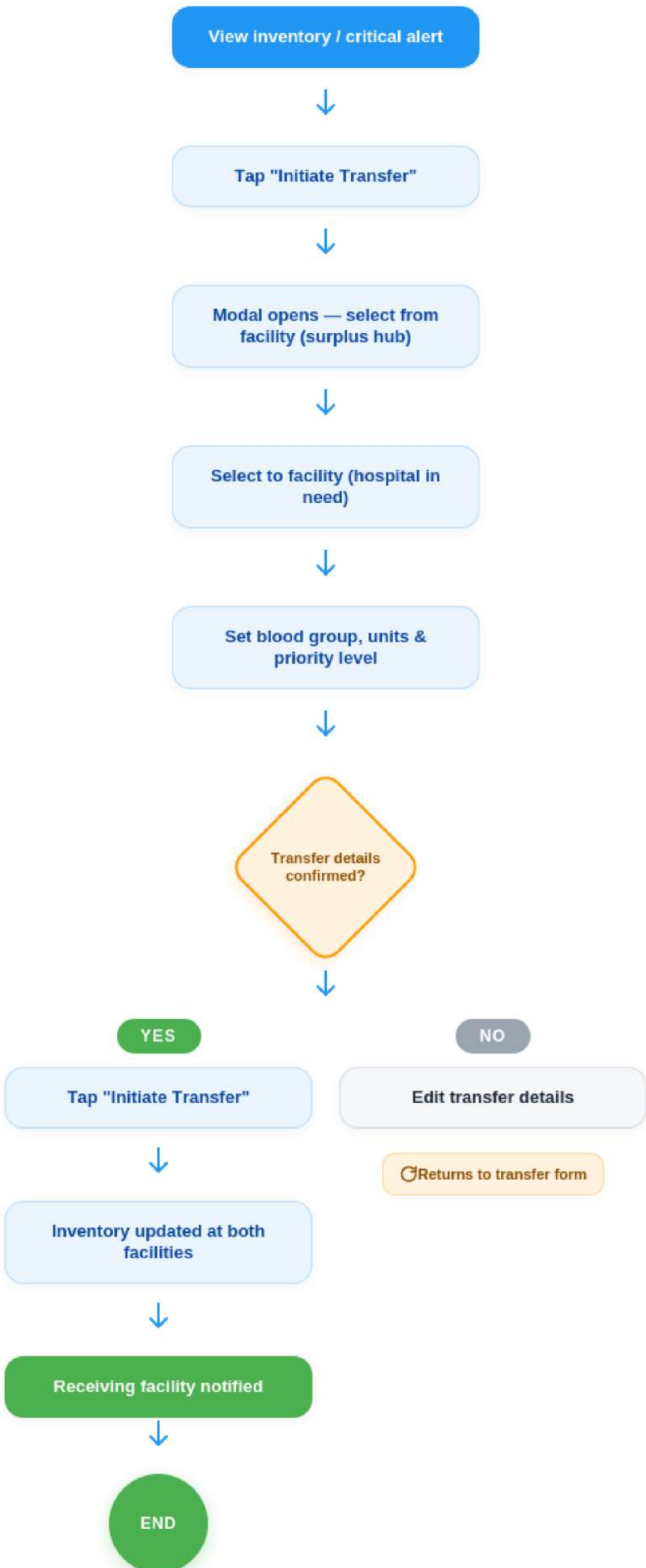
2 Contact top donors flow

How Dr. Sharma responds to a critical stock alert by reaching AI-recommended donors.



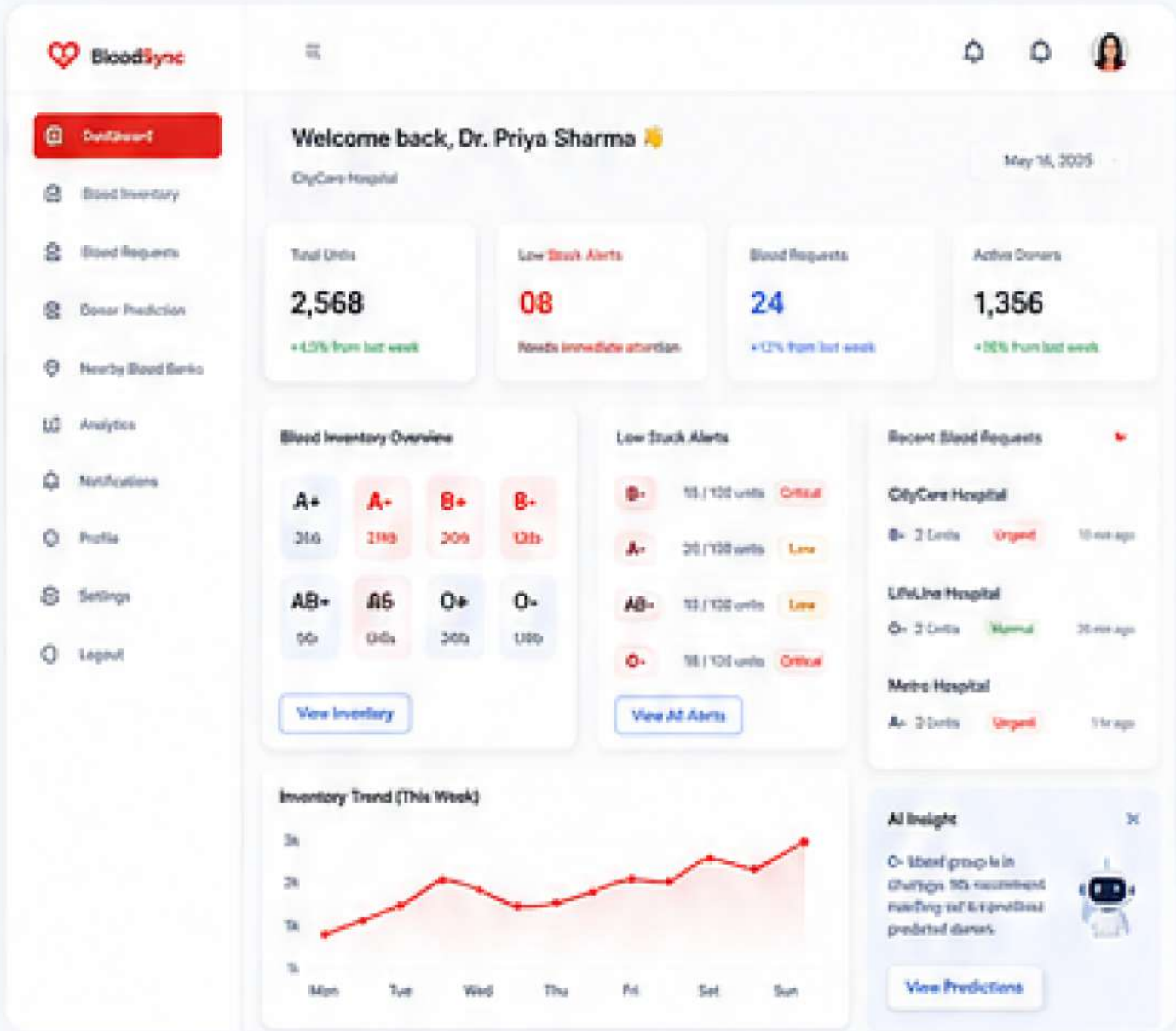
3 Initiate transfer flow

How Dr. Sharma moves surplus units from one facility to a facility in shortage.



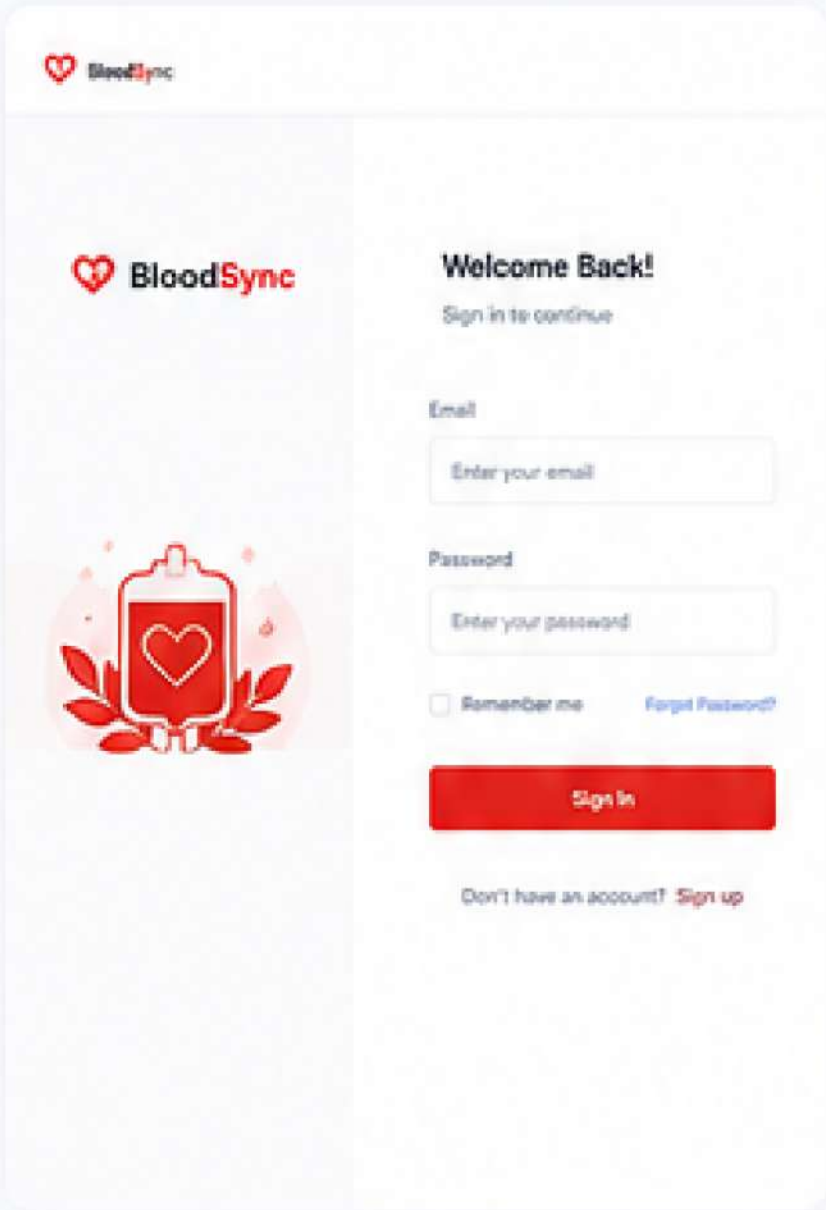
Final UI Screens

A cohesive set of high-fidelity screens designed to simplify blood bank operations through clear dashboards, predictive insights, real-time inventory monitoring, and streamlined workflows.



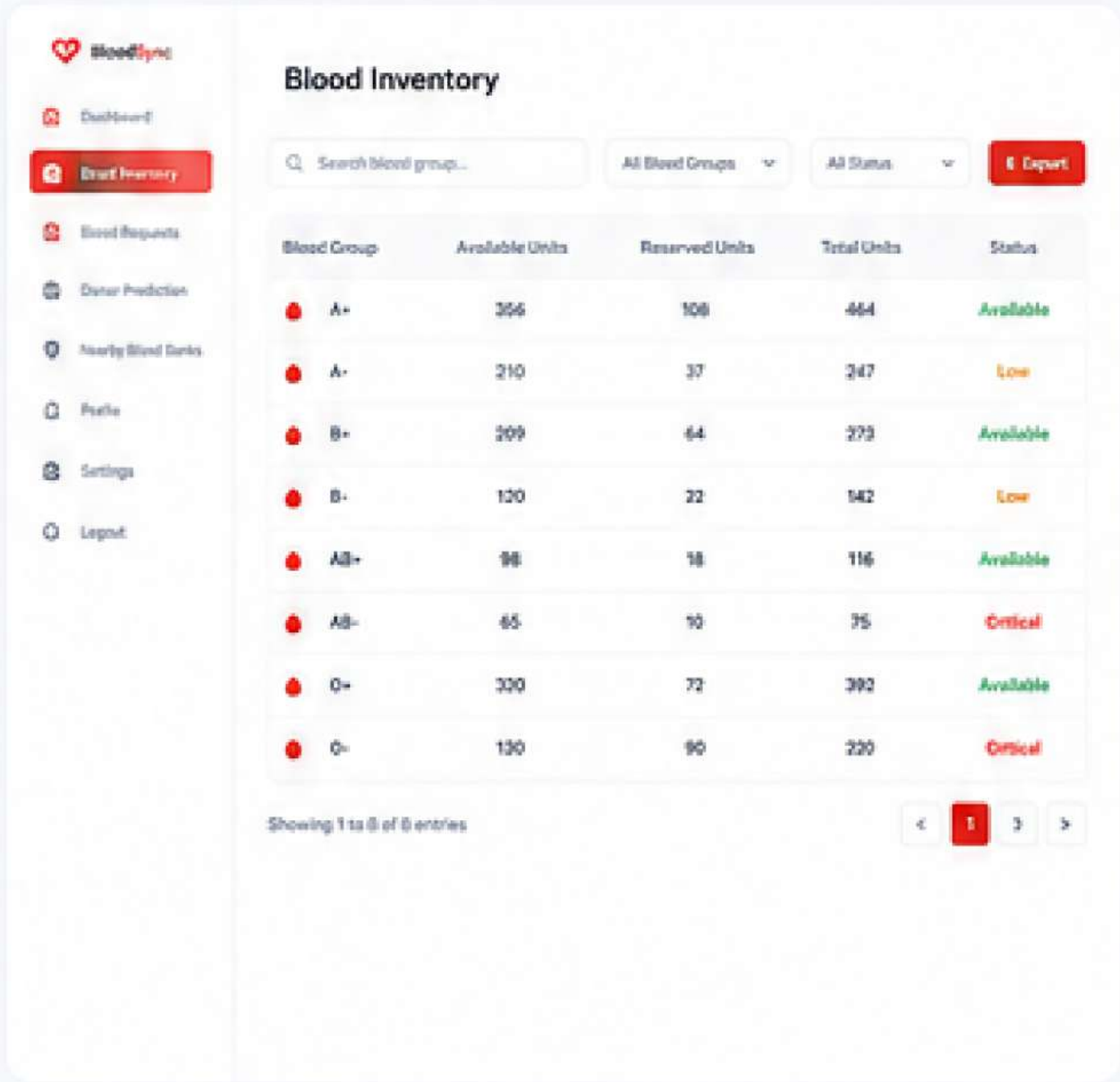
01 Hospital Dashboard

An overview of key metrics, inventory status, alerts, requests and insights.



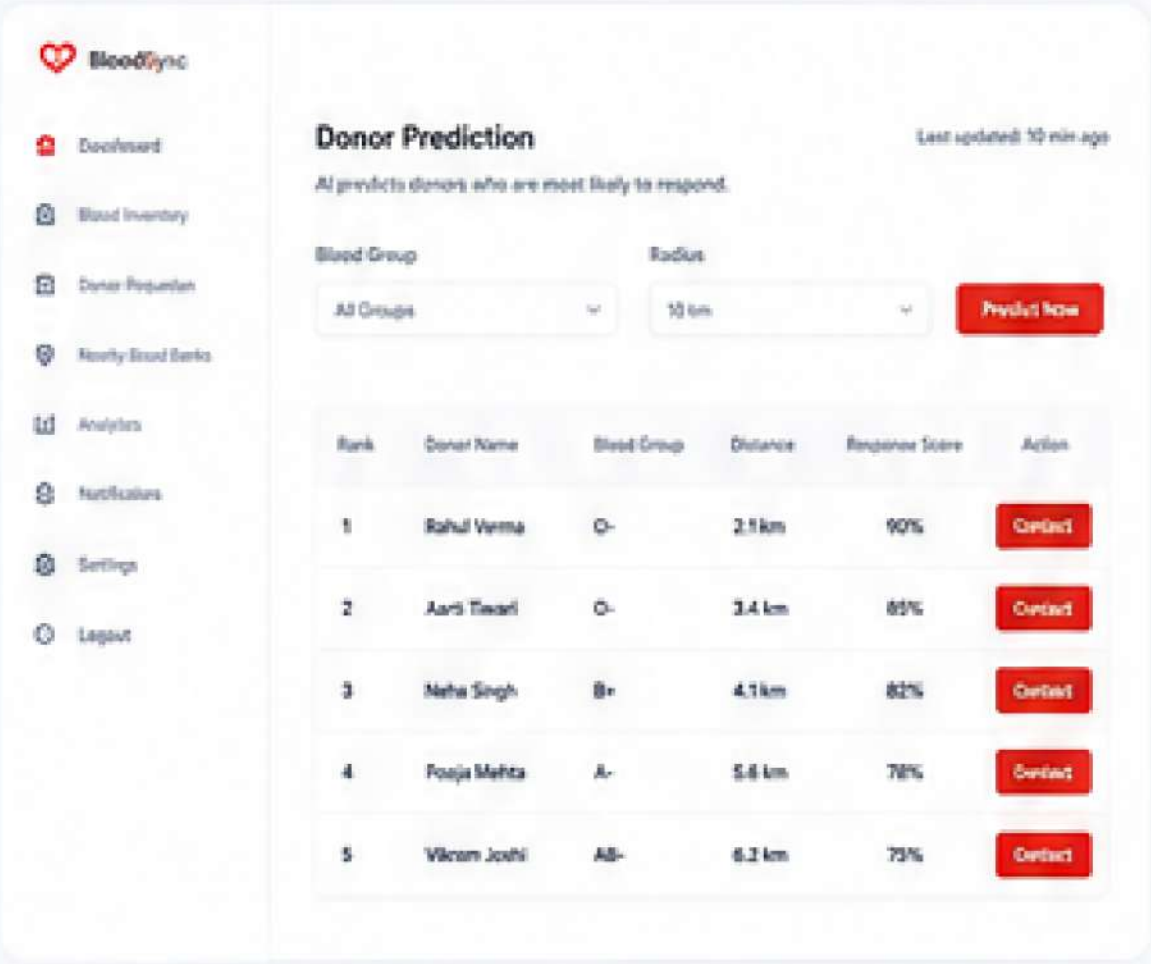
02 Login Screen

Secure login screen for hospitals and blood banks.



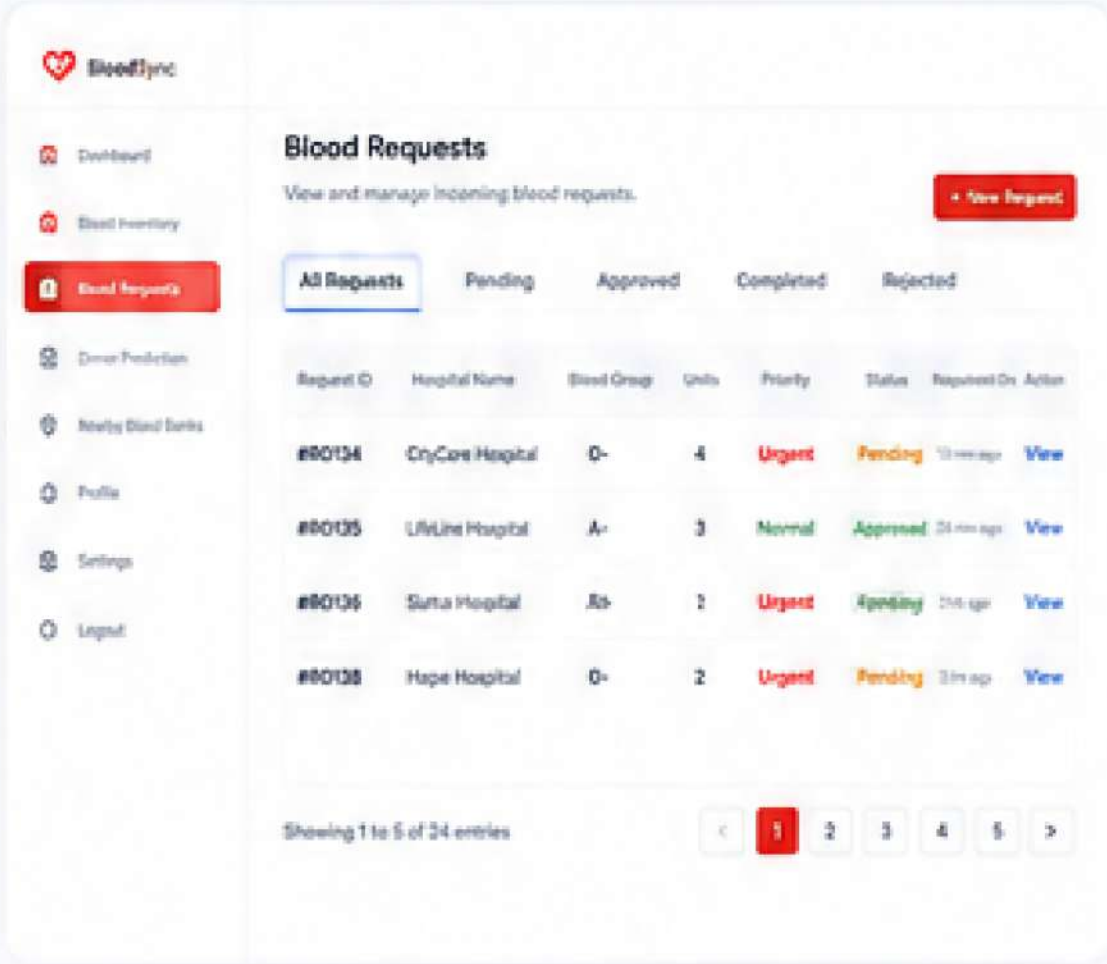
03 Blood Inventory

View and manage blood stock levels in real-time.



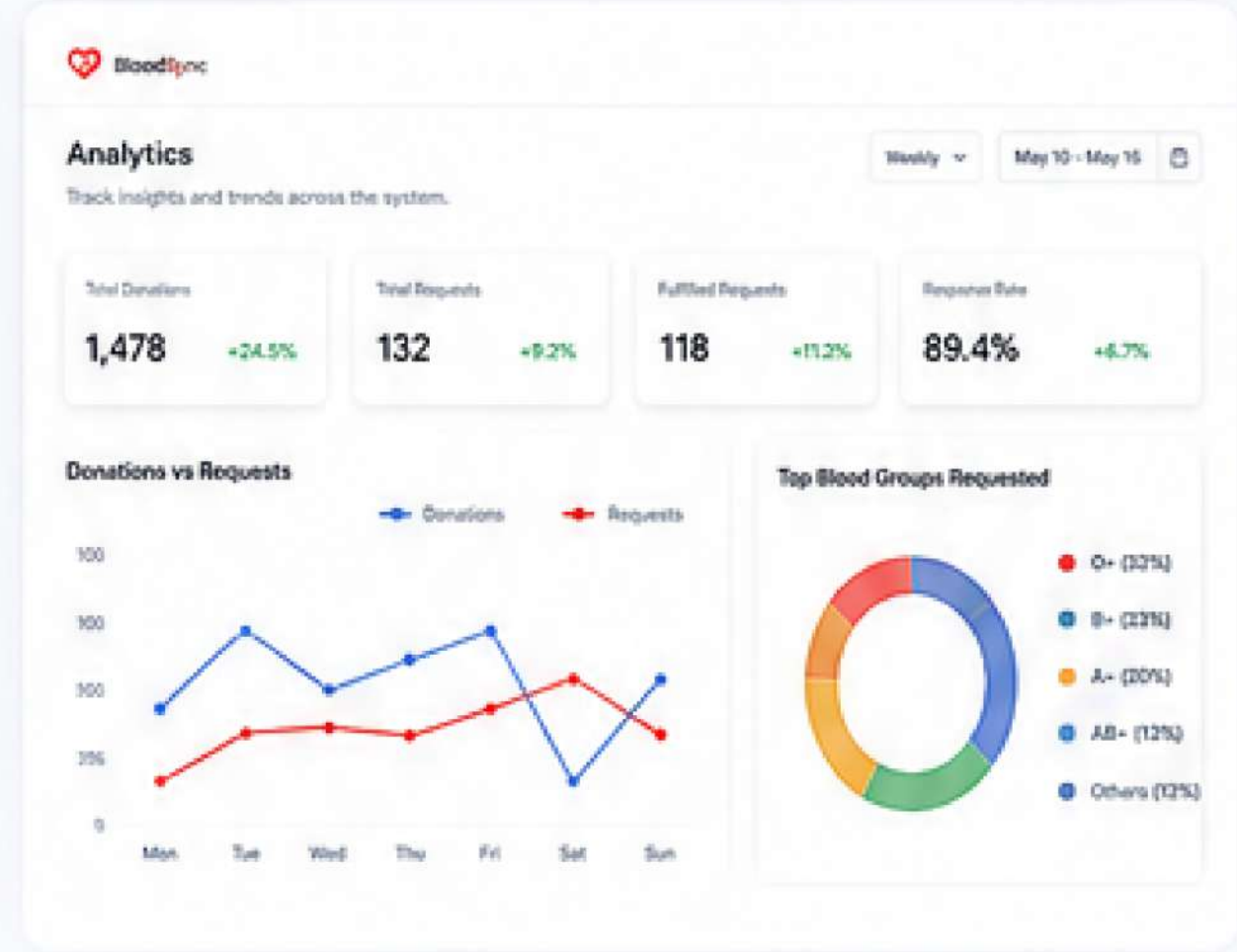
04 Donor Prediction

AI-powered predictions to identify likely donor responders.



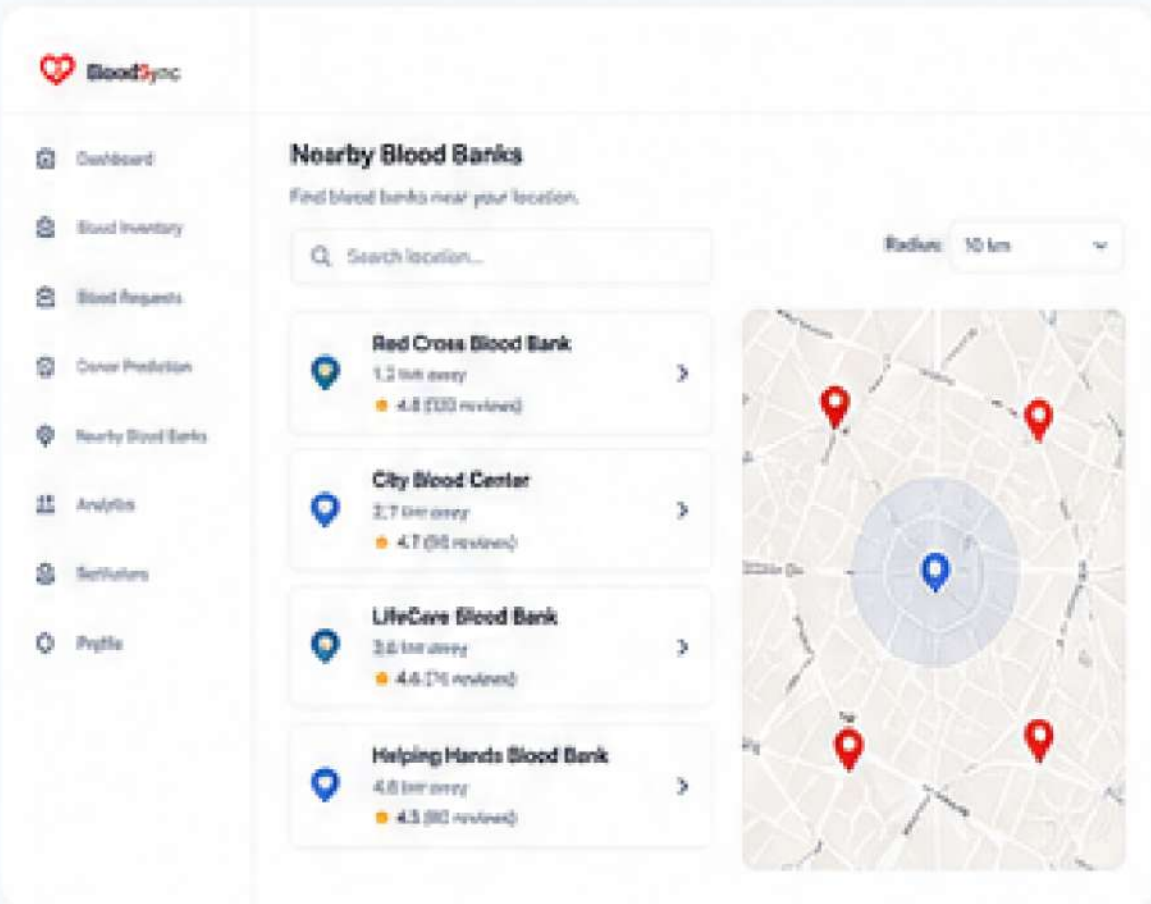
05 Blood Requests

Track and manage blood requests from different hospitals.



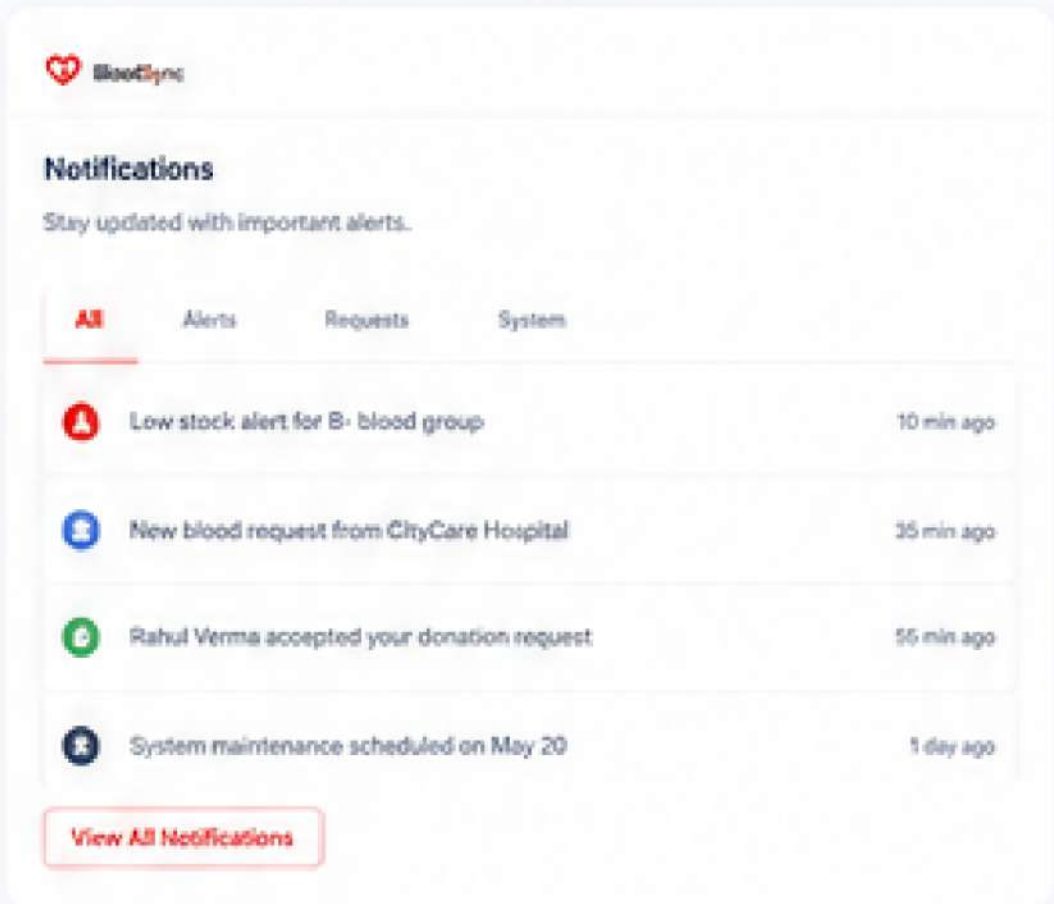
06 Analytics

Visual insights on donations, requests, response rate and trends.



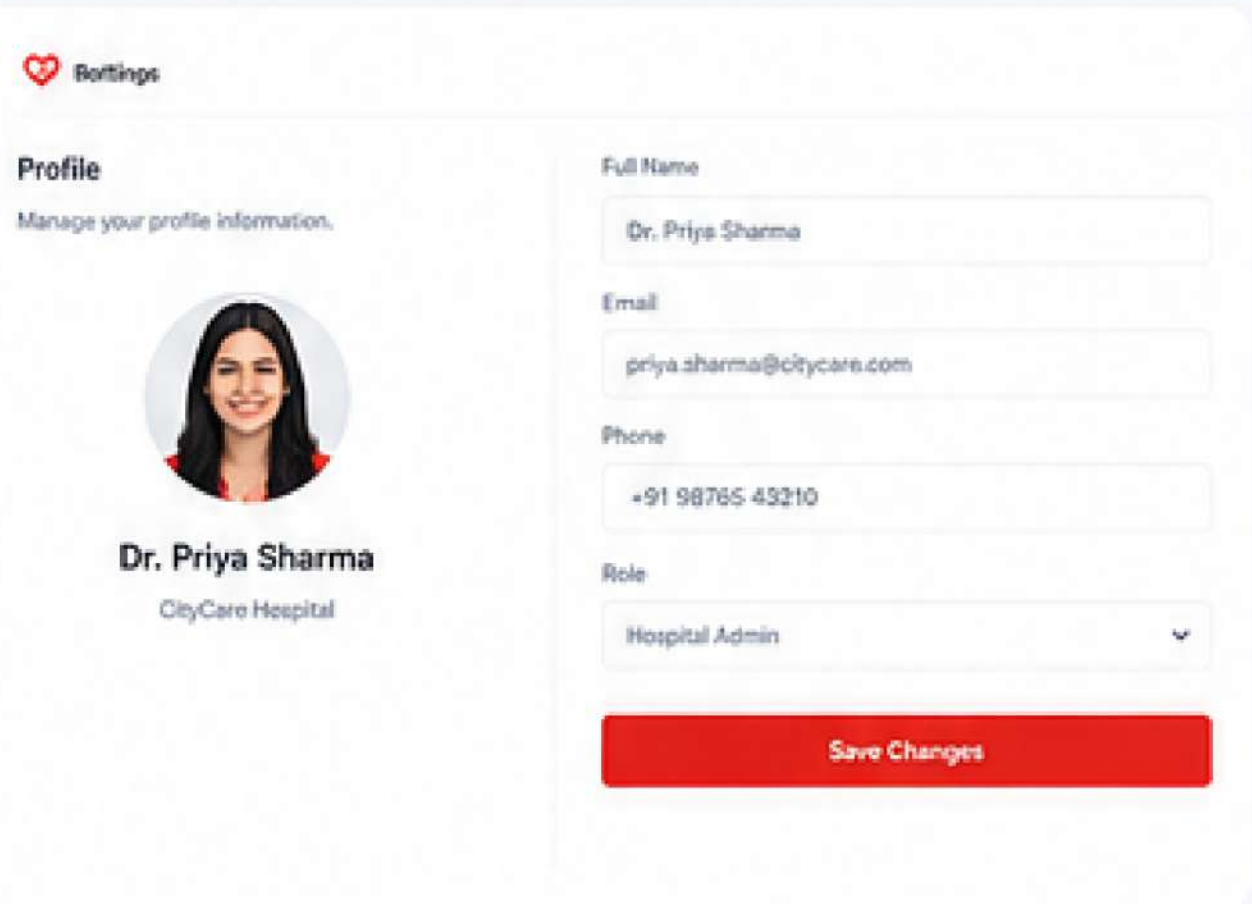
07 Nearby Blood Banks

Locate nearby blood banks and view distance and ratings.



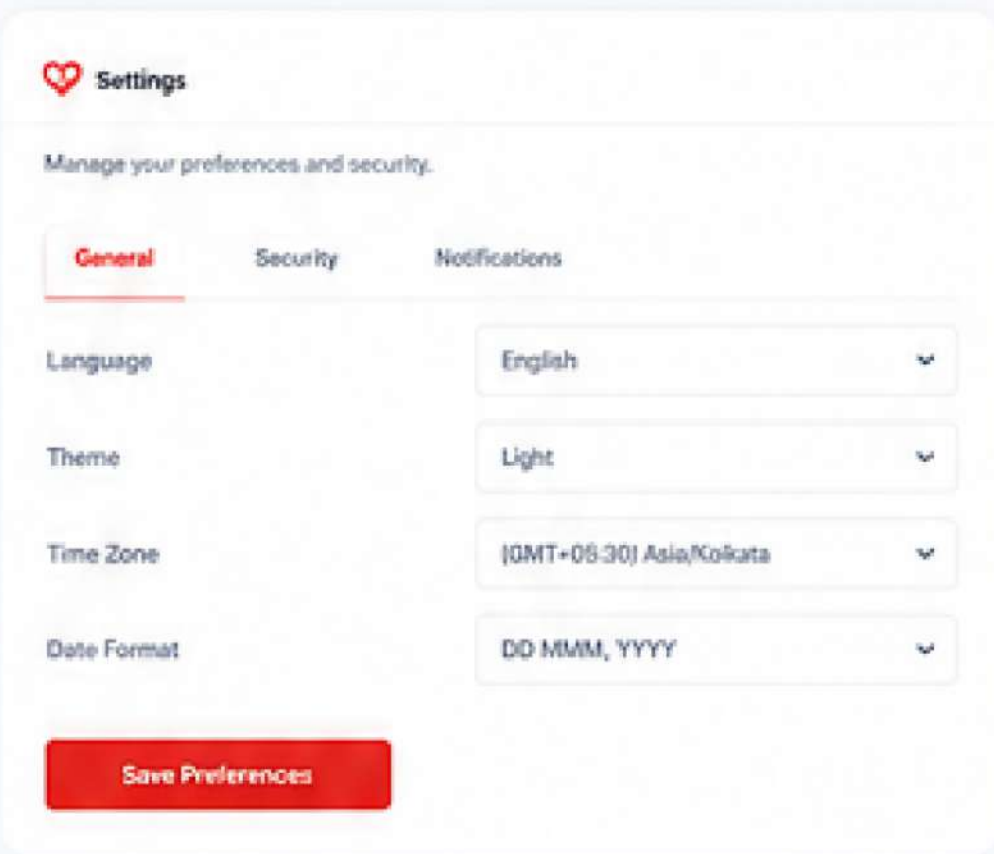
08 Notifications

Stay informed with alerts, requests and system updates.



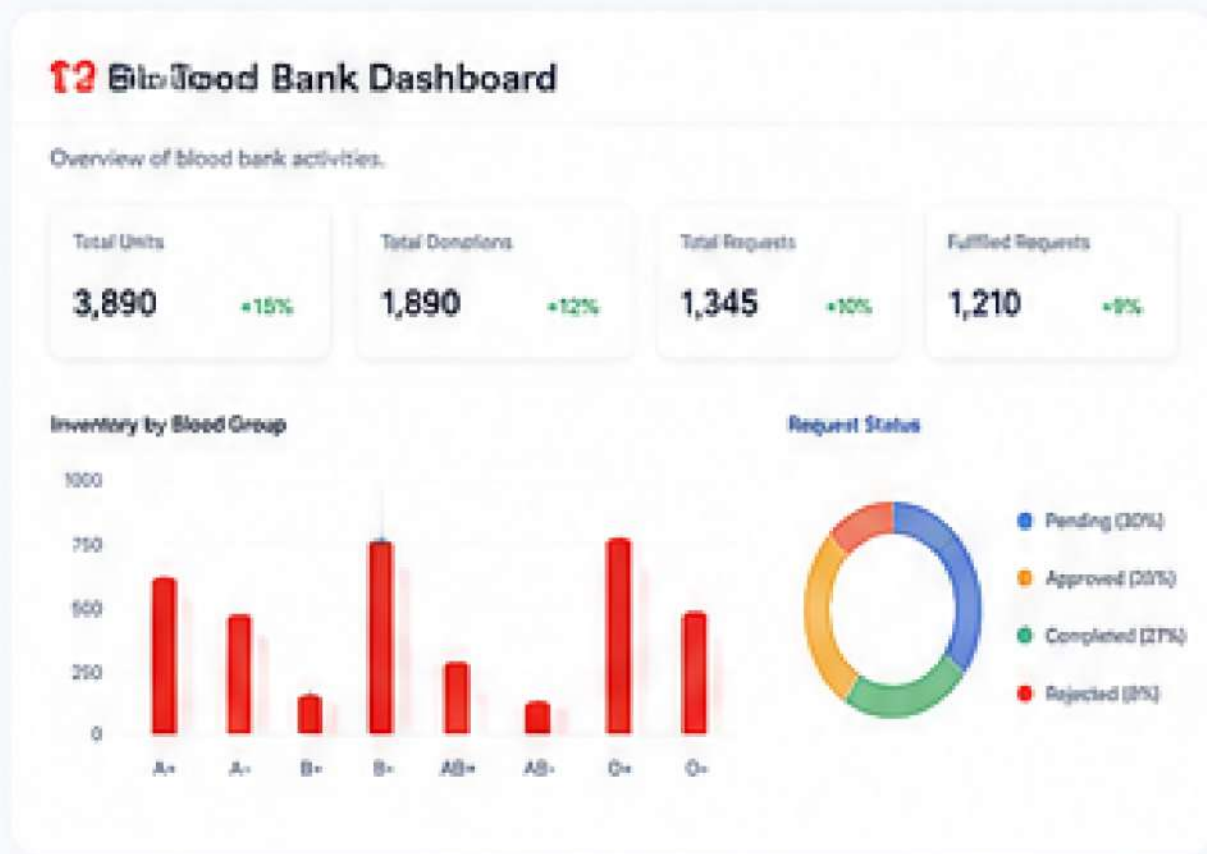
09 Profile

View and update your personal and hospital information.



10 Settings

Customize app preferences and security settings.



11 Blood Bank Dashboard

Overview of inventory, requests and activity for blood banks.



12 System Overview

High-level overview of platform performance and growth.

Thanks for watching
